

ORIGINAL ARTICLE

A dual-channel robust deep learning framework for enhanced detection of hardware Trojans via side-channel analysis

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Abstract

The detection of Hardware Trojans (HTs) in electronic components is critical to ensuring the security and integrity of electronic systems, as these malicious modifications can leak sensitive information, alter device functionality, or completely disable the device. This study introduces a novel detection method combining deep learning techniques with dual-channel image transformations to identify HTs via Side-Channel Analysis (SCA). Specifically, our approach converts side-channel time series signals, such as power consumption, electromagnetic emissions, and timing data, into dual-channel, image-like representations using Gramian Angular Field (GAF) and reshaping techniques. These transformations allow convolutional neural networks (CNNs), known for their effectiveness in image analysis, to capture subtle and complex anomalous patterns indicative of HTs. We rigorously evaluated our dual-channel methodology using publicly available Advanced Encryption Standard (AES) datasets for HTs provided by TrustHub and IEEE Dataport. Experimental results demonstrate that the proposed approach achieves superior accuracy compared to existing methods, particularly in challenging datasets. Additionally, we assessed the robustness of our model by introducing varying noise levels to simulate real-world operational perturbations such as process variations, aging, and voltage fluctuations. The proposed method maintains high detection accuracy and demonstrates enhanced resilience under noisy conditions, underscoring its practical applicability and effectiveness in detecting sophisticated HT threats.

Keywords Hardware Trojan · Dual-channel time series imaging · Gramian angular field transformation · Convolutional neural network · Deep learning · Side-channel analysis

1 Introduction

Hardware Trojans (HTs) are malicious modifications to electronic circuits or designs that can cause incorrect behavior of an integrated circuit (IC) [1]. These Trojans can be stealthily integrated into a chip's design or manufacturing process, making their detection challenging. They are engineered to stay dormant, activating under specific conditions to avoid discovery and maintain normal operation in most scenarios [2]. The concept of HTs has gained importance with the globalization of the semiconductor supply chain [3]. As electronic components production and design processes become more globalized and complex, the opportunity for malicious entities



to introduce unauthorized alterations increases. These alterations can be as subtle as a few extra transistors or a small change in the wiring of a chip, yet their impact can be profound. HTs can have disastrous consequences. Once activated, they can disrupt the normal operations of a system, leading to incorrect results, leaking confidential information, or providing backdoor access to sensitive systems. In critical infrastructures, such as financial systems, healthcare equipment, or national security systems, the activation of an HT could lead to catastrophic outcomes, including loss of data, financial damage, or even threats to human safety [4].

The sophistication of HT attacks allows them to evade traditional testing and verification methods. Given their stealthy nature and the potential for significant harm, HTs have become a central concern for cybersecurity experts. The increasing threat of HTs highlights the critical need for robust security measures to protect hardware from such vulnerabilities. Detecting HTs and mitigating their effect are critical for developing next-generation defensive mechanisms for electronics development and deployment in the presence of the HT threat. Therefore, continued research and development in this area are imperative in order to prevent the potentially devastating consequences of HTs and to maintain the integrity of the technology infrastructure [5]. One primary method for the detection of HT attacks is Side-Channel Analysis (SCA). This technique involves the monitoring of physical parameters like power consumption, electromagnetic emissions, timing intervals, and even acoustic signals while the hardware is in operation [6]. Detecting malicious alterations to hardware components can be effectively accomplished by analyzing these parameters, which deviate from expected norms in the presence of a Trojan. Techniques like power and electromagnetic analysis can detect anomalies indicative of HTs. However, these methods may encounter difficulties with the noise and variations in normal operation, leading to false positives or negatives [7].

Integrating machine learning with SCA can improve the detection of HTs substantially [8–10]. In this combined approach, machine learning algorithms are employed to process and analyze side-channel signals generated by the electronic devices. These algorithms create sophisticated mathematical models that capture the standard physical and operational behaviors of the devices. This modeling enables the effective identification and categorization of irregularities and anomalies in the side-channel signals, which are often indicative of HTs. Machine learning in this context improves Trojan detection accuracy and provides adaptability in response to continuously evolving hardware threats.

While traditional machine learning techniques such as Multilayer Perceptron (MLP) can process side-channel signals to detect HTs, they may have difficulty dealing with the high dimensionality and noise present in these signals. This can lead to less accurate detection [11]. Utilizing Convolutional Neural Networks (CNNs) for HTs detection offers significant advantages. CNNs, as a form of deep learning, are very efficient at detecting subtle and complex patterns within large and noisy datasets [12]. By effectively learning from the detailed characteristics of side-channel signals, CNNs can enhance the sensitivity and accuracy of HTs detection. Their ability to filter out irrelevant noise and focus on important features makes them particularly adept at reducing false alarms. Moreover, the sophisticated algorithms employed in CNN architectures allow for a more precise analysis of side-channel data, leading to more reliable identification of hardware anomalies potentially caused by Trojans. However, while 1D CNNs have been widely used for processing time series data directly, for 2D CNNs to be effectively utilized on side-channel time series data, the data needs to be transformed into image-like formats. This transformation allows CNNs to use their inherent strengths in spatial pattern recognition, making them a more powerful and reliable tool in hardware security.

The Gramian Angular Field (GAF) transformation plays a crucial role in enhancing the capability of CNNs [13, 14]. By converting time series data into a matrix format that visually resembles an image, GAF effectively addresses the disparity between temporal data and the image-centric processing strength of CNNs [13]. This transformation not only preserves the temporal correlation between different time points in the series, which is essential for accurate HTs detection, but also formats the data in a way that enhances the pattern recognition capabilities of CNNs. The image-like representation created by GAF can reveal a variety of features, such as cyclic patterns and hidden trends, that are often less discernible in the original time series format [15]. The ability of CNNs to analyze images can then be utilized to identify subtle anomalies and patterns indicative of HTs.

This ability to learn complex patterns from GAF-transformed data potentially leads to enhanced performance in identifying HTs, a task that requires identifying subtle and often obscure discrepancies in hardware behavior. Therefore, the integration of GAF with CNNs can make significant advancements in the field of hardware security, particularly in the sophisticated and critical domain of HTs detection.

This paper's primary contribution is the introduction of a novel method for detecting HTs through a dual-channel deep learning architecture, using SCA and time-series imaging. A key innovation of our approach is the application of dual-channel transformations utilizing the GAF and reshaping techniques to convert time-series data into a format conducive to complex pattern recognition by CNNs. To the best of our knowledge, this technique has not previously been used for HT detection. This dual-channel transformation process helps in recognizing complex patterns, significantly improving the accuracy of HT detection. Furthermore, we extensively evaluate the robustness of our proposed detection system against various real-world operational perturbations, such as noise introduced by process variations, aging, and voltage level changes. Our findings demonstrate that our method maintains high accuracy and robustness, making it highly suitable for practical applications where such conditions are prevalent. Additionally, the impact of noise on the discriminability of signals across different benchmarks is investigated, along with an analysis of how the addition of noise to test sets in certain benchmark datasets affects the performance of HT detection. Comprehensive simulations, using publicly available datasets, have been conducted to assess the performance of the proposed network relative to existing methods. The results of this comparative analysis underscore the superior efficacy of the proposed dual-channel approach in detecting HT attacks.

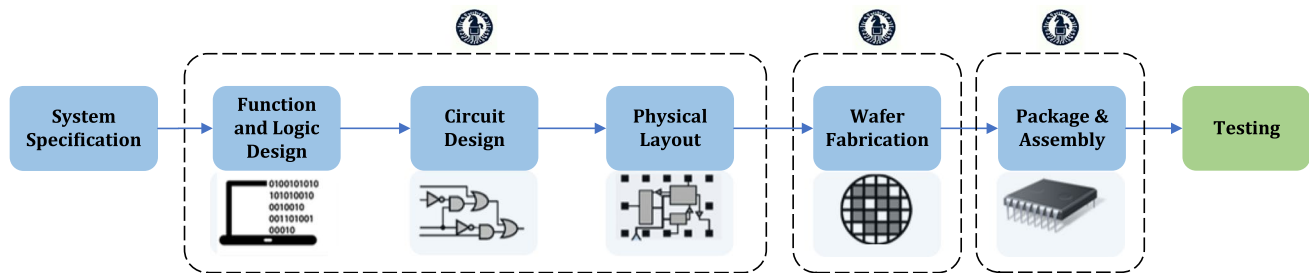
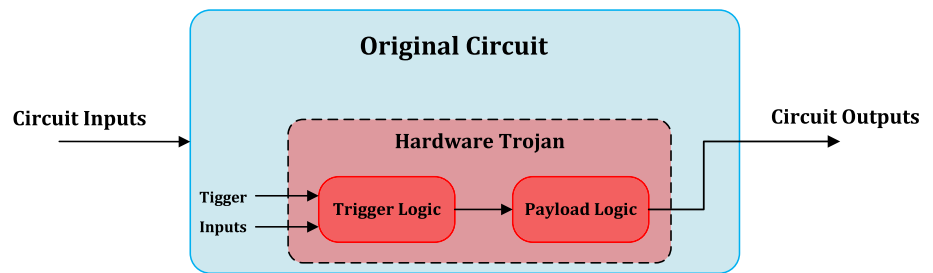
The structure of this article is organized as follows. Section 2 presents the background on HT attacks and provides an overview of existing HT detection methods. In Sect. 3, the proposed methodology is outlined, with a focus on imaging time series data and the application of GAF. The results and discussion are elaborated in Sect. 4, where the effectiveness of the proposed approach is analyzed in detail. Finally, Sect. 5 concludes the paper with a summary of the key findings and implications of this study. To provide a foundation for our proposed method, the next section reviews key background concepts and summarizes previous approaches to Hardware Trojan detection.

2 Background and related works

Understanding HT attacks is crucial due to their serious threat to the security and reliability of electronic systems, including critical infrastructure and consumer electronics, as they pose significant risks that could lead to catastrophic failures and breaches of privacy. Consequently, developing effective detection methods is essential for identifying and neutralizing these Trojans before they can cause harm. This will ensure hardware systems' reliability and security in an increasingly interconnected world. This section provides a comprehensive background essential for understanding the context and challenges associated with HT attacks and the various methods developed for their detection.

2.1 Hardware Trojan attack

HTs can be categorized according to their activation triggers and the nature of their malicious actions, or payloads [16]. The trigger mechanism is the condition or set of conditions that activate the HT. These triggers can be highly specific, designed to remain dormant and undetectable until a particular event occurs. Triggers vary widely, from simple conditions like a specific date or time to complex patterns of activity or specific inputs that might only occur under rare circumstances. The sophistication of the trigger mechanism helps the Trojan evade detection during routine testing and operational use, ensuring it remains inactive until it can achieve its intended effect. Once activated, the payload of an HT is the actual malicious action carried out by the Trojan. The payload can range from altering the function of the device, leaking sensitive information to an unauthorized party, degrading

Fig. 1 General configuration of a hardware Trojan [17]**Fig. 2** An overview of IC design and manufacturing process, highlighting potential points for HT insertion

the system's performance, or even physically damaging the hardware. The nature of the payload is dictated by the attacker's objectives, ranging from simple disruptions to severe threats that compromise national security by targeting critical infrastructure. Figure 1 presents a simplified block diagram of an embedded HT, illustrating the trigger and payload mechanism.

As shown in Fig. 2, HTs can be embedded at various stages of a chip's lifecycle, taking advantage of vulnerabilities from the initial design phase to the fabrication phase, and extending to the final deployment stage [4]. In the design phase, these malicious entities can be stealthily introduced by either tampering with third-party Intellectual Property (IP) cores, which are often incorporated into chips to save on development time and costs, or by compromising the software tools used for chip design. This could involve the insertion of malicious code or circuitry that remains dormant until triggered under specific conditions. The fabrication phase is also vulnerable when manufacturing is outsourced, as untrusted facilities may modify chip layouts or alter masks used in lithography, particularly when the manufacturing process is outsourced to facilities across the globe. This global outsourcing model, while economically advantageous, allows the untrusted facilities to subtly modify chip layouts or tamper with the masks used in the lithography process, thereby embedding HTs directly into the silicon. Such alterations are incredibly challenging to detect, given the microscopic scale and complexity of modern integrated circuits. In the assembly and packaging stages, additional vulnerabilities can emerge. These stages provide yet another opportunity for the insertion of HTs, either through the direct modification of the silicon die itself or through the inclusion of malicious components within the chip package. Furthermore, as chips are distributed through the supply chain to reach assemblers or end-users, there exist opportunities for interception and subsequent tampering, adding another layer of risk.

The complex nature of these threats underscores the critical importance of vigilance and advanced security measures at each stage of a chip's lifecycle. To minimize the risk of HTs and protect the security and functionality of electronic systems, a comprehensive approach is essential. This includes applying secure and verified design techniques, ensuring the integrity of manufacturing and assembly processes, and securing the supply chain. A critical aspect of HT detection, however, is the development and implementation of sophisticated detection methods. These techniques are crucial for identifying and neutralizing Trojans that might evade initial preventive measures, ensuring the ongoing security and functionality of electronic systems.

2.2 Related works

Detecting HTs remains a critical challenge due to their stealthy nature and their ability to mimic normal circuit behavior [18]. Over the years, detection methods have evolved and can be broadly divided into two categories: pre-silicon and post-silicon approaches [19, 20]. Pre-silicon methods emphasize secure design and verification techniques to preempt Trojan insertion [21], whereas post-silicon methods rely on techniques such as functional testing, logical analysis, and Side-Channel Analysis (SCA) to identify HTs after chip fabrication.

Methods for detecting HT in post-silicon processes can generally be divided into two categories, destructive and non-destructive [21]. Destructive methods for identifying Trojans involve physically deconstructing the device and using invasive techniques such as reverse engineering to inspect its internal structure and components. This not only increases the cost, due to the need for multiple samples, but also limits the feasibility of widespread testing, especially in large-scale production. Non-destructive methods, such as logic testing and SCA, are commonly used because they are designed to examine the device without causing any physical damage [22]. Logic testing involves verifying the logical integrity of the IC by checking its functionality against its intended design [23]. The idea is to stimulate the chip with a range of input conditions and observe the outputs to detect any anomalous behavior. The limitation of logic testing is its inability to detect dormant Trojans that are designed to be triggered under very specific or rare conditions [24]. Such Trojans might remain undetected under standard test conditions. In addition to these hardware-based test strategies, several design-for-trust approaches have been proposed, which integrate security features directly into the chip design. While these methods can prevent the insertion of Trojans or facilitate early detection, they often introduce additional overhead and require continuous updates to address emerging threats [25].

SCA detection capability in HT detection is significant due to its non-intrusive nature and the ability to uncover extremely stealthy Trojans that might remain dormant or undetectable under conventional testing [26]. It plays a vital role in ensuring hardware integrity and security in a wide range of applications, from consumer electronics to critical infrastructure systems. Its importance has grown with the increasing complexity of modern electronic devices and the global nature of electronics supply chains, where the risk of tampering or unauthorized alterations is heightened. SCA involves monitoring unintended signals of the device's operation, such as power consumption, electromagnetic emissions, and timing information, to detect abnormalities that may indicate the presence of a Trojan [6]. SCA is particularly effective because it can potentially detect Trojans that are inactive during standard logic testing [27]. By analyzing the device's behavior under different conditions and comparing it to an established normal baseline from a golden chip, potential anomalies indicative of an HT can be identified. This method is particularly effective at detecting stealthy Trojans designed to evade conventional logic tests, which makes it an essential component of modern hardware security protocols. However, traditional SCA methods may encounter problems with noise and variations in normal operation, leading to false positives or negatives [16, 28].

Integrating machine learning with SCA substantially enhances the detection of HTs [29, 30]. In this integrated methodology, machine learning techniques are utilized to interpret side-channel data from electronic devices. These techniques develop complex mathematical representations that capture devices' normal operational patterns. Through this representation, it becomes feasible to accurately detect and classify deviations within the side-channel emissions, which frequently signal the presence of HTs. Employing machine learning for this purpose increases the accuracy in identifying Trojans [31]. Recent review papers have demonstrated the critical role of machine learning methods in the detection and security of HTs. A comprehensive survey is provided in [8] on utilizing machine learning for analyzing various hardware security vulnerabilities, illustrating its effectiveness in detecting complex issues and underscoring the importance of integrating machine learning into prevention and mitigation strategies for future security enhancements. An extensive review of HT attacks, highlighting the crucial role of machine learning in detection and the necessity of design-for-security principles in protecting against the HT threats is presented in [31]. The review in [32] studies the HT attacks, illustrating the critical role of machine learning in enhancing detection through both traditional and advanced neural network-based methods. The study in [33] examines machine learning's application to identifying and countering HTs within IoT devices,

highlighting specific features like power and temperature for detection. The survey study in [23] analyzes methods for detecting and preventing HTs in electronic systems, emphasizing the potential of machine learning for feature extraction and classification in the absence of a golden reference, and exploring physical parameter measurement and logic testing techniques for Trojan detection.

Extensive research has been conducted on the detection of HT attacks using side-channel signals with different machine learning algorithms. These algorithms have proven effective in identifying subtle discrepancies in power consumption, electromagnetic emissions, timing differences, and other physical parameters that are indicative of malicious alterations within hardware components. Through the application of techniques such as support vector machines, neural networks, and decision trees, researchers have enhanced HT detection methods' sensitivity and accuracy. Numerous research efforts have explored a range of machine learning algorithms through both supervised and unsupervised techniques to tackle the subtle and sophisticated nature of HT attacks [20, 34]. Advanced methodologies have been utilized, including Support Vector Machine (SVM) [35], Naive Bayes [36], Random Forest [37], and XGBoost classifiers [38]. These methods exploit complex patterns in side-channel signals, such as power consumption and electromagnetic emissions, to detect anomalies that signify HT attacks. Moreover, innovative techniques like Deep Belief Networks (DBN) optimized with Stacked Autoencoder Hashing Optimization (SAEHO) [39], Siamese neural networks [40], Contrastive Learning [41] and attention mechanism-enhanced models [42], represent state-of-the-art approaches that enhance detection accuracy and adaptability in the field of HT attack identification using side-channel analysis. The combination of SCA and CNN offers improved feature extraction and robustness to noise, making it a powerful tool for detecting Hardware Trojans and other security threats. Unlike traditional methods, SCA-CNN can handle high-dimensional data efficiently and reduces the need for manual feature engineering, providing a scalable and effective solution [43, 44]. A key design principle in multi-representation learning is representational diversity, where complementary views of the same data are combined to capture richer feature spaces and improve robustness. In our case, the dual-channel architecture integrates the GAF branch, which captures global temporal correlations, with the reshaped-raw branch, which preserves local temporal structure and amplitude variations. This diversity strengthens generalization and resilience to noise. These algorithms represent the state of the art in applying machine learning to HT detection, addressing the evolving challenges of modern cybersecurity threats.

These efforts demonstrate an evolving approach to dealing with hardware security threats, using the capabilities of machine learning and deep learning to develop sophisticated and increasingly effective detection mechanisms. While significant progress has been made in using SCA and machine learning for HT detection, challenges remain in handling noise, high-dimensional data, and stealthy Trojans. To address these challenges, we propose a novel deep learning-based framework that transforms side-channel signals into image-like representations using GAF. The next section details our proposed method and dataset used for evaluation.

Building on the insights from the literature, the following section presents the dataset, outlines the proposed detection pipeline, and details the dual-channel imaging methodology.

3 Hardware Trojan detection pipeline

This section first discusses the dataset used in this study, which consists of power side-channel signals associated with both disabled and triggered HTs. It provides essential background information and explains why this dataset was chosen. We will describe the dataset in detail, including its specific attributes and characteristics. In the second part, we will present the implementation of our proposed methodology, designed to detect and analyze potential HT threats.

3.1 Overview of dataset

The dataset utilized in this study originates from the publicly accessible IEEE DataPort website [45], as outlined by [46]. It consists of power side-channel signals from a subset of HTs referenced by Trust-Hub [47], specifically targeting a cryptographic core circuit, the Advanced Encryption Standard (AES) with a 128-bit key length, which is a cryptographic algorithm used to protect sensitive data. The AES circuit, upon receiving a 128-bit plaintext input, encrypts it with a secret key to produce a 128-bit ciphertext output. For data collection, the encryption process is executed 10,000 times, generating an equal number of time-series signals with a size of 2500 [45].

According to [46], the data is collected using two Sakura-G boards, focusing on an AES circuit to capture power consumption signals that indicate the presence or activation of HTs. This dataset contains data gathered under various conditions, including when the HT is disabled or actively triggered. The dataset's uniqueness is in its coverage of both static and dynamic components of power consumption across different operational states of HTs. This provides a rich basis for developing and evaluating HT detection models. This detailed dataset enables the development of a statistical model aimed at improving HT detection methods' accuracy and robustness.

The ideal power consumption of an AES circuit within a golden chip consists of two parts of dynamic power and static power [48]:

$$P_T = P_D + P_S \quad (1)$$

Here, P_T represents the overall intrinsic power consumption, comprising both the dynamic power, denoted by P_D , and the static power, represented by P_S . However, power measurements can be affected by noise due to process variations, supply voltage fluctuations, temperature changes, and measurement limitations [49]. The observed power consumption is thus given by:

$$\tilde{P}_T = P_T + P_N \quad (2)$$

where P_N is noise power. After activation of an HT, additional power consumption is introduced to the circuit. This extra consumption changes the observed power data and can be expressed as:

$$\mathcal{P}_T = \tilde{P}_T + P_{TR} \quad (3)$$

where P_{TR} represents the power added by the triggered Trojan. By analyzing and comparing the power signals $\mathcal{P}_T$ and $\tilde{P}_T$, it is feasible to detect the presence of an active HT compared to its dormant state. However, one of the significant challenges in side-channel-based HT detection is the presence of noise. The HT power consumption signature can easily be obscured by noise. This problem of noise can mask the additional power consumed by an HT, greatly reducing the accuracy of side-channel-based attack detection methods. Consequently, distinguishing between the normal power consumption of an AES circuit and the altered consumption due to an HT becomes a complex task.

In this paper, we focus on five specific Trojan types for the sake of comparison with previous works, out of the twelve documented in the available dataset. These are AES-T500, AES-T600, AES-T700, AES-T800, and AES-T1600. These benchmarks describe various scenarios in which an HT can be triggered and the subsequent impacts on the AES circuit [46]. The AES-T500 is activated by a specific sequence of numbers and then drains the battery through a perpetually rotating shift register. The AES-T600 triggers upon receiving a predefined number and then leaks the secret key by incrementing the leakage current for each 0 bit within the key. Similarly, both the AES-T700 and AES-T800 are activated by a certain number and sequence of numbers, respectively, and both leak the key through a covert channel using a CDMA sequence code. Lastly, the AES-T1600, which also responds to a specific sequence of numbers, leaks the key via an RF signal, potentially allowing remote eavesdropping. Each benchmark presents a unique combination of trigger conditions and payloads, demonstrating the diverse

and insidious nature of HT attacks on AES. Figure 3 illustrates two sample power signals for disabled and triggered Trojan states.

From the dataset, which includes 10,000 time-series signals with the disabled Trojan and another 10,000 with the triggered Trojan, we allocate 80% of the signals from each set are allocated for the training set and 20% for the validation set, ensuring robust training and validation for HT detection in AES circuits.

3.2 Proposed dual-channel time-series imaging-based detection method

This section discusses the proposed deep learning method for detecting HT attacks. The method utilizes reshaping techniques and the GAF transformation to convert time-series data into images, creating a two-channel image. This innovative approach enhances the model's ability to detect Trojan attacks through side-channel signals, demonstrating its proficiency in analyzing complex data structures to pinpoint security vulnerabilities. Figure 5 illustrates the process of visualizing time series data through the GAF transformation, demonstrating three key stages, the original time series data, its conversion via polar transformation, and the final GAF representation. This progression highlights the method's capability to transform time series into a visually interpretable format, facilitating the detection of patterns and anomalies within the data.

Subsequently, these images undergo processing through a series of CNN blocks within the model's architecture, as presented in Fig. 6. Each CNN block in the model applies a set of filters to the input image to extract features. This feature extraction is enhanced through batch normalization, ensuring stability and efficiency in the training process. Following the convolutional operation, the model employs a non-linear transformation, utilizing a ReLU function. This step introduces non-linearity to the model, allowing it to learn more complex patterns in the data. The transformed data is then passed through a pooling layer, which reduces the spatial dimensions (downsizing the output) while retaining the most important features. This downsizing is crucial for reducing the computational load and preventing overfitting by abstracting the features. The sequential arrangement of CNN and pooling layers is repeated across the model, ensuring thorough feature extraction and representation for accurate classification.

Then, the output from the model's convolutional layers proceeds through a flattening step, transforming the multi-dimensional feature maps into a one-dimensional vector. This flattened vector then feeds into a series of densely connected (fully connected) layers. The final step of this process is a Softmax classifier, which assigns the predicted category, indicative of whether an HT attack is triggered or not. During training, the main aim is to minimize the error between the actual label of the input data and the model's output, with cross-entropy loss being a standard error metric for classification tasks. The deep learning model structure is presented in Fig. 6.

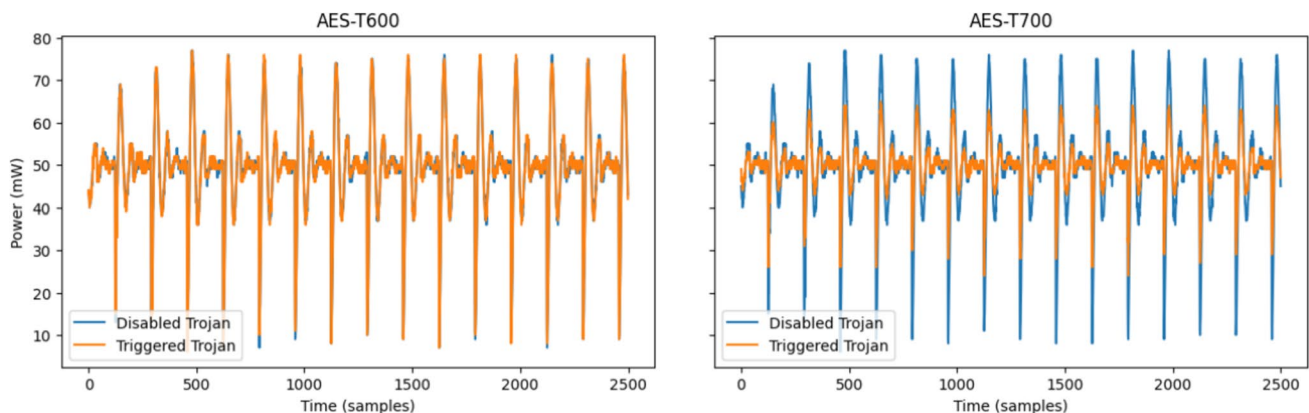


Fig. 3 Two sample signals of disabled and triggered Trojan of AES-T600 (left) and AES-T700 (right)

In the subsequent section, the process of transforming time series data into image-like formats is detailed. As illustrated in Fig. 5, this transformation effectively prepares the data for input into a CNN model.

3.3 Image-based time series data transformation

Transforming time series data into image-like formats has shown significant performance enhancements in diverse fields, enabling the application of CNNs for effective time series analysis by using their strength in spatial feature recognition [50–52]. On the other hand, processes such as AES involve repetitive computations that are cyclic in nature. During these computations, certain operations are repeated each cycle, which can lead to quasi-periodic patterns in power consumption [53, 54]. Periodic or quasi-periodic signals can be arranged in the form of a 2-D array. This arrangement allows the correlations among the data to be effectively exploited with a 2-D-based compression scheme. For example, in medical imaging, such as ECG signal analysis, it is common to convert 1-D time-series data into a 2-D spectrogram or similar representation. This transformation aids in better visualization and analysis of frequency components over time [55].

One method for transforming time-series data into an image-like format involves organizing the linear sequence of data points into rows and columns to form an image, as illustrated in Fig. 4. Another popular method for this transformation is the use of GAF imaging. GAF imaging is a technique that converts time series data into images, allowing CNNs to identify visual patterns for analysis [56]. To this goal, GAF method transforms data points into angles and distances, with cosine values depicting angular differences [57]. This transformation of data points into angles and distances allows complex temporal patterns of time-series signals to be visualized and analyzed. The GAF approach is particularly advantageous as it translates temporal changes into spatial patterns, which facilitates the use of two-dimensional CNNs [13]. These CNNs are adept at interpreting spatial relationships and patterns within the data, thereby enhancing the model’s ability to uncover and analyze latent features and intricate patterns. The use of GAF imaging has been proposed in several studies, demonstrating its effectiveness in different domains, such as fault detection, financial forecasting and stock market analysis [58–60].

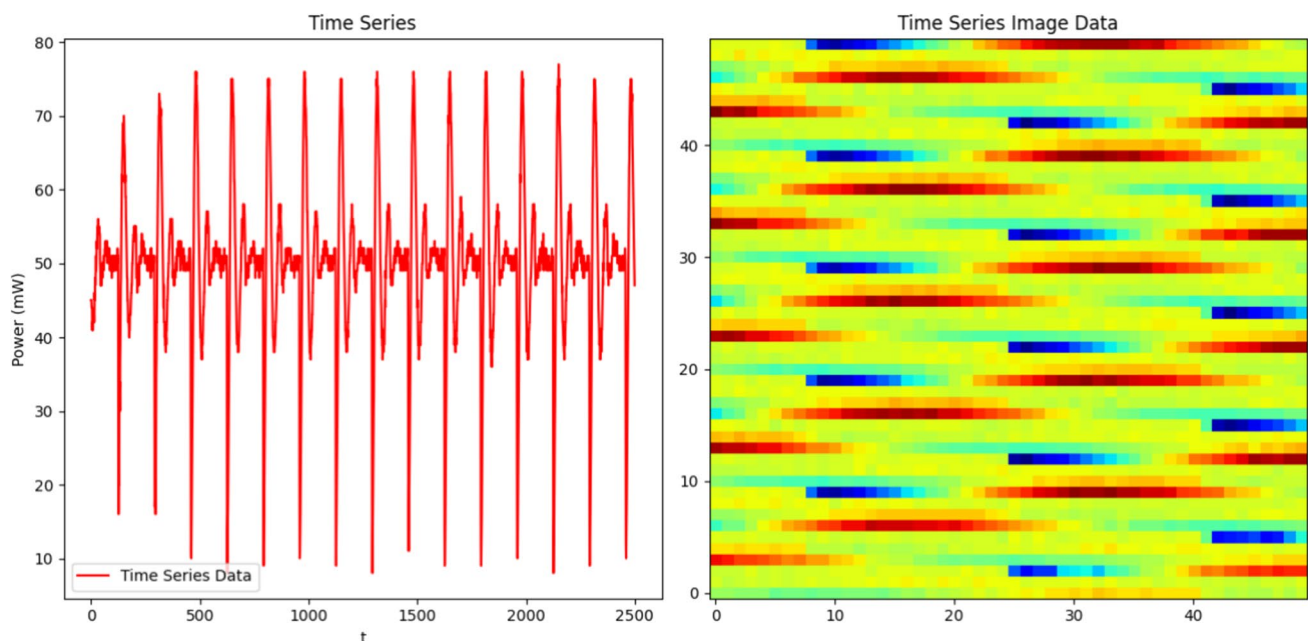


Fig. 4 Visualizing Time Series with reshaping: (Left) Original Time Series, (Right) Reshaped Image Representation

3.4 Implementing GAF for time series data

GAF represents the time series in a polar coordinate system as shown in Fig. 5. The process involves several mathematical steps. First, the time series data is normalized to scale the values to fit within the trigonometric function limits, typically to the range $[-1, 1]$. Given a time series $\mathcal{T}_s = \{x_1, x_2, \dots, x_n\}$, the normalized series $\hat{\mathcal{T}}_s$ is given by:

$$\hat{\mathcal{T}}_s = 2 \times \frac{\mathcal{T}_s - \min(\mathcal{T}_s)}{\max(\mathcal{T}_s) - \min(\mathcal{T}_s)} - 1 \quad (4)$$

In the next step, each normalized value $\hat{x}_i$ is transformed into a polar coordinate. The value $\hat{x}_i$ represents the cosine of the angular component, while the time stamp t_i is used as the radial component:

$$\theta_i = \arccos(\hat{x}_i), \quad \text{for } i = 1, 2, \dots, n \quad (5)$$

$$r_i = \frac{t_i}{\mathcal{R}}, \quad t_i \in \mathbb{N} \quad (6)$$

where θ_i is the angular component, r_i is the radial component, t_i is the time stamp and $\mathcal{R}$ serves as a constant regularization factor to maintain the span of the polar coordinate space.

Then, the GAF is computed by taking the outer product of the angular component. The Gramian Angular Summation Field (GASF) or Gramian Angular Difference Field (GADF) is obtained depending on whether the cosine of the sum or difference of angles is used. The element S_{ij} of GASF is computed as:

$$S_{ij} = \cos(\theta_i + \theta_j) \quad (7)$$

Therefore, the GASF can be computed as the following matrix:

$$S = \begin{pmatrix} \cos(\theta_1 + \theta_1) & \dots & \cos(\theta_1 + \theta_n) \\ \cos(\theta_2 + \theta_1) & \dots & \cos(\theta_2 + \theta_n) \\ \vdots & \ddots & \vdots \\ \cos(\theta_n + \theta_1) & \dots & \cos(\theta_n + \theta_n) \end{pmatrix} \quad (8)$$

The element of i, j the matrix D for GADF, on the other hand, is computed as:

$$D_{ij} = \sin(\theta_i - \theta_j) \quad (9)$$

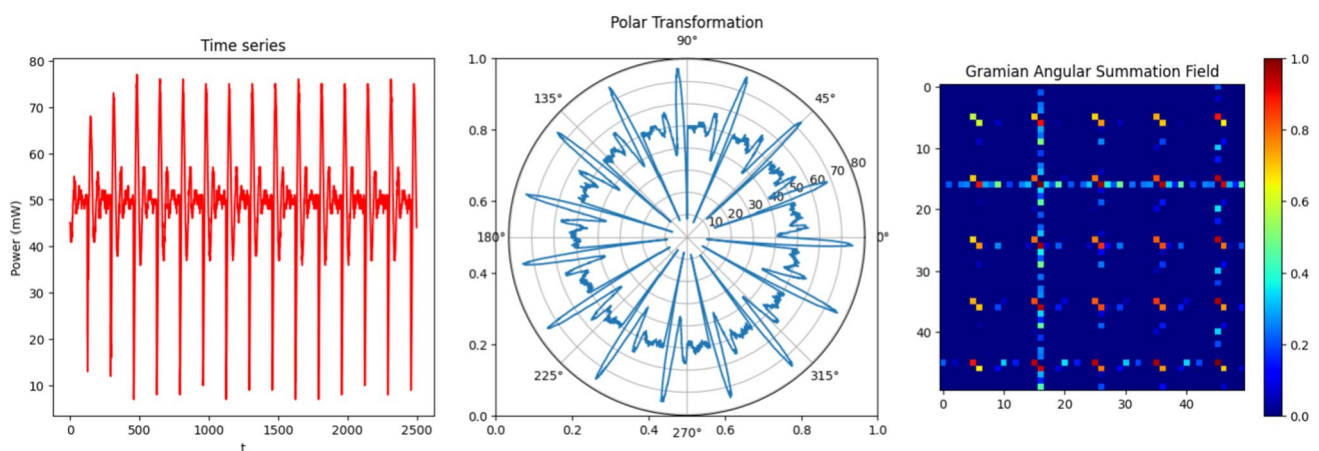


Fig. 5 Visualizing Time Series with GAF: (Left) Original Time Series, (Center) Polar Transformation, (Right) GAF Representation

Accordingly, the GADF can be computed as the following matrix:

$$D = \begin{pmatrix} \sin(\theta_1 - \theta_1) & \dots & \sin(\theta_1 - \theta_n) \\ \sin(\theta_2 - \theta_1) & \dots & \sin(\theta_2 - \theta_n) \\ \vdots & \ddots & \vdots \\ \sin(\theta_n - \theta_1) & \dots & \sin(\theta_n - \theta_n) \end{pmatrix} \quad (10)$$

Both matrices S for GASF and D for GADF are $n \times n$ matrices, where n is the length of time series. GASF or GADF matrix represents the time series data in a 2D image-like format, capturing the relative temporal correlation between different times in the series. This transformation effectively encodes temporal patterns into a 2D spatial format, suitable for processing by CNNs for tasks such as classification, regression, or anomaly detection. The GAF transformation preserves temporal relationships and can uncover hidden patterns in the time series. Figure 7 depicts the imaging of power signals for various HT benchmarks, contrasting the disabled Trojan with the triggered Trojan, as processed by the GASF Transform.

In our implementation, we apply the GASF variant. Each time-series signal is first resampled or zero-padded to a fixed length of 64 samples. We apply min-max normalization to map the signal to the range $[-1, 1]$, as required for the angular encoding step. The GAF transformation then computes the cosine of pairwise angular sums, resulting in a symmetric image that preserves temporal dependencies. The resulting 64x64 GAF image is used as one channel, while the reshaped raw signal (also resized to 64x64) serves as the second channel. This dual-channel image is then fed into the CNN. These choices ensure consistent input dimensions and enhance reproducibility.

To adjust the size of the n -sized windows of time-series data to either a smaller or larger size, several methods can be applied [61]. One common technique is resampling, which involves either downsampling to reduce the size or upsampling to expand the size of the data windows. Downsampling reduces data frequency by aggregating multiple data points into one, effectively decreasing the n window size. Conversely, upsampling increases frequency by interpolating additional data points between existing ones, and thus increases the n window size. Therefore, to modify the output image dimensions, the time series length is initially adjusted to match the desired image dimension.

Remark 1 As shown in Figs. 4 and 5, these transformations create a dual-channel image. The dual-channel approach allows different aspects of the data to be highlighted and processed. While the reshaped time series might help the CNN identify abrupt changes and anomalies directly from raw data patterns, the GAF channel provides insights into the cyclic nature and deep temporal relationships. This means the network can simultaneously learn from both raw and transformed representations, leading to a more comprehensive feature extraction process.

With the dataset, preprocessing, and model architecture defined, we now evaluate the proposed framework's performance and robustness under various experimental scenarios in next section.

4 Results and discussions

In this section, results and discussion are provided about our novel time series imaging-based detection method. First, the performance of our proposed HT detection method is evaluated using the dataset discussed earlier. Then, the robustness of our method is assessed against noise in the validation dataset, a critical factor for evaluating the method's real-world applicability. Finally, the impact of noise on the discriminability between disabled and triggered Trojan signals is discussed. This is essential for assessing the effectiveness of the proposed HT detection method.

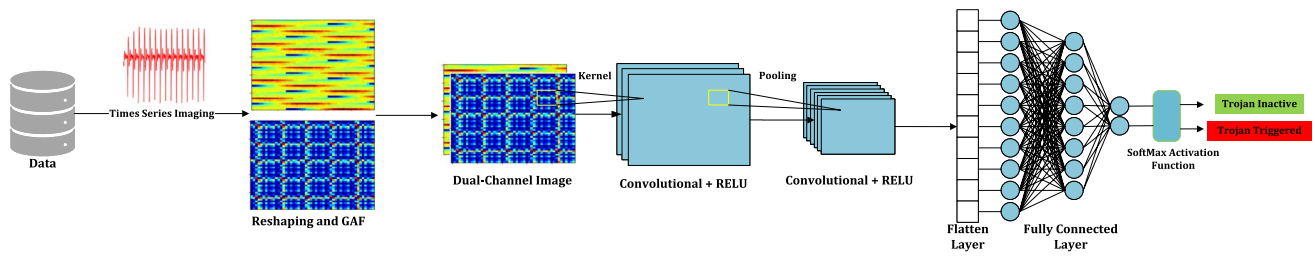


Fig. 6 Overview of the model for classification of inactive and triggered Trojan attacks based on side-channel signal analysis

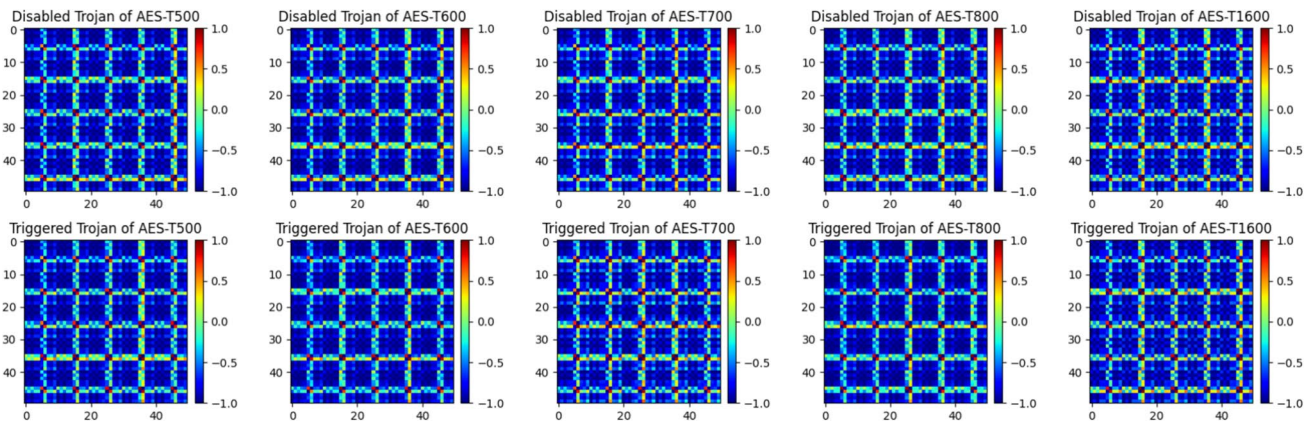


Fig. 7 Gramian Angular Field Transformation of Side-Channel Signals: (Top) Comparing Triggered Trojan vs. (Bottom) Disabled Trojan Signals

Table 1 CNN model architecture

Layer type	Filters/nodes	Kernel size	Stride	Padding
Convolutional Layer 1	32	3	1	2
Max Pooling Layer	—	3	2	—
Convolutional Layer 2	64	3	1	2
Fully Connected Layer 1	200	—	—	—
Fully Connected Layer 2	2	—	—	—

4.1 Evaluation of the proposed HT detection method

For the purpose of simulation and evaluation of our proposed time series imaging-based learning method, a selection is made where 80% of the signals are allocated for the training set and 20% for the validation set. In our study, we explore the application of the model, as depicted in Fig. 6 and implemented using PyTorch version 2.1.2, to accurately detect HT attacks.

The model's initial analysis begins with two convolutional layers dedicated to feature extraction. The first layer employs 32 filters, each with a kernel size of 3, a stride of 1, and padding of 2. This configuration allows the model to capture the initial feature set from input. A subsequent max pooling layer, with a kernel size of 3 and a stride of 2, reduces the spatial dimensions of the feature maps, keeping only the most important information. To enhance the analysis depth, the second convolutional layer utilizes 64 filters, maintaining the same kernel size, stride, and padding to further refine the extracted features. The model then proceeds into a classification phase with two fully connected layers. The first layer maps the flattened feature maps into a 200-dimensional space, while the second condenses this data into a 2-dimensional output, aligning with the classification targets. Table 1 presents the parameters of the CNN model.

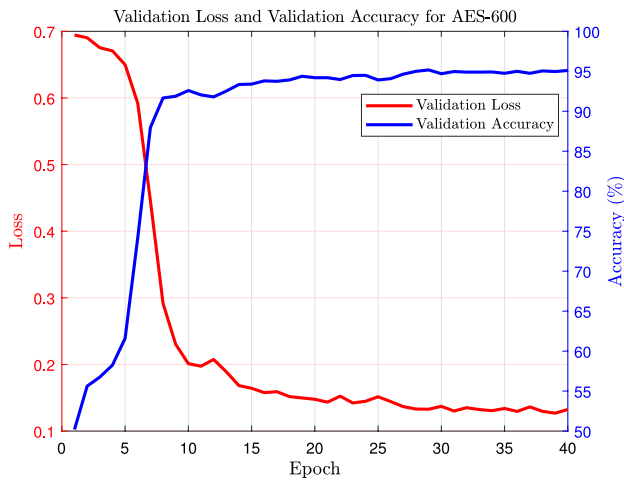


Fig. 8 Validation loss and accuracy for AES-600

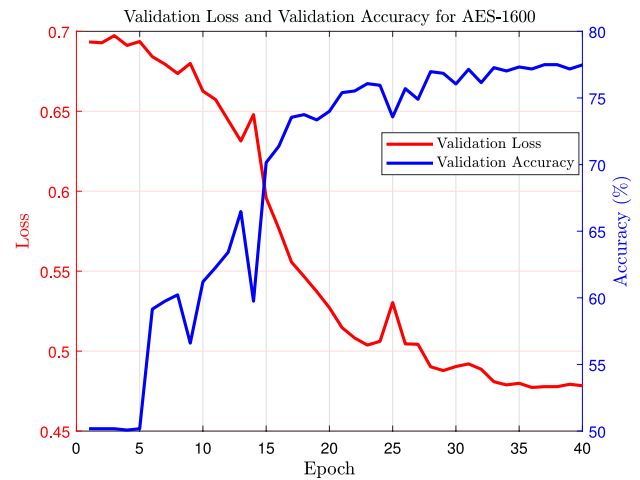


Fig. 9 Validation loss and accuracy for AES-1600

The performance of the proposed model for detecting HTs in AES encryption modules for the AES-600 and AES-1600 HT benchmarks is demonstrated in Figs. 8 and 9, which show the model's validation accuracy over training epochs. The loss (red curve) and accuracy (blue curve) metrics depicted in the Figs. 8 and 9 are indicators of the model's performance in classifying HTs within the benchmarks.

In this work, the loss is calculated using the cross-entropy loss function for binary classification, given by the following equation:

$$L(y, \hat{y}) = -\frac{1}{N} \sum_{i=1}^N [y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)] \quad (11)$$

where y represents the true label, $\hat{y}$ the predicted probability of the positive class, and N the number of samples in the dataset. Here, y_i indicates the presence or absence of an HT, and $\hat{y}_i$ is the predicted probability that the i -th sample contains an HT.

Accuracy, on the other hand, is computed as follows:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}(\hat{y}_i = y_i) \quad (12)$$

with $\mathbb{1}(\cdot)$ being the indicator function that is 1 when $\hat{y}_i$ is equal to y_i and 0 otherwise. Together, these metrics provide a comprehensive view of the model's classification ability.

To evaluate the performance of our proposed methodologies for HT detection, we conducted a comprehensive comparison of our approach with several established methods previously evaluated in [46]. Specifically, our evaluation includes the One-Class Support Vector Machine (OCSVM), recognized for its robust performance in anomaly detection, along with three neural network architectures: MLP, CNN, and Hierarchical Temporal Memory (HTM). HTM draws inspiration from the human neocortex structure to effectively process temporal data. The performance metrics for these established models are directly obtained from the findings reported in [46]. In addition to these, we implemented two raw-signal-based baselines: a RawTrace-CNN designed to capture local temporal patterns directly from normalized power traces, and a Long Short-Term Memory (LSTM) network that models sequential dependencies through its gated memory cell structure. Furthermore, we compare our method against single-channel approaches based solely on Markov Transition Field (MTF) and GAF transformations. The

Table 2 Comparison of results with different methods on each AES dataset (0.0 to 1.0 scale)

Method	AES-T500	AES-T600	AES-T700	AES-T800	AES-T1600
CNN	0.845	0.555	1.000	1.000	0.640
MLP	0.850	0.655	1.000	1.000	0.705
LSTM	0.983	0.618	1.000	0.999	0.543
RawTrace-CNN	0.980	0.932	1.000	1.000	0.687
MTF	0.980	0.793	1.000	1.000	0.561
GAF	0.985	0.585	0.921	0.945	0.523
OC-SVM	0.897	0.497	0.902	0.927	0.502
HTM [46]	1.000	0.635	0.981	1.000	0.596
ARA [43]	1.000	0.673	1.000	1.000	0.631
Our Method	0.985	0.952	1.000	1.000	0.775

MTF method converts raw time series into image representations by encoding the transition probabilities among discretized signal states, thereby capturing the dynamic signal evolution [57]. Both the single-channel MTF and GAF approaches utilize the identical CNN architecture outlined in Table 1. Further extending the scope of our comparison, we also include the cutting-edge Attention-Res-Attention (ARA) network, as reported in [43]. This approach utilizes the ResNeXt architecture combined with an attention mechanism to focus on salient features that may indicate the presence of HTs. In contrast to these one-channel methods, our primary contribution is a dual-channel approach that uses the GAF transformation together with a raw signal reshaping technique, yielding a more robust feature extraction process and superior detection performance, particularly under noisy conditions.

The comprehensive results are depicted in Table 2, which presents a comparative analysis of the detection rates across different models. As shown in the figure, our time-series imaging-based detector achieves superior performance on AES-T600 and AES-T1600, and comparable performance on AES-T500, AES-T700, and AES-T800. The accuracy for AES-600 is about 95.2% and for AES-T1600 is about 77.5%, both of which are higher than the accuracy reported by previous studies. The AES-T1600 dataset is particularly challenging for HT detection due to its complex signal patterns and subtle variations as shown in the next subsection. Despite these difficulties, our proposed method achieves an accuracy of 77.5%, which is a notable accomplishment compared to the reported accuracy of 59.6% in [46] and 63.1% in [43]. This result demonstrates the effectiveness of our approach in addressing the difficulties of detecting HTs in challenging datasets. This indicates of our model's robust feature extraction capabilities and its ability to generalize from complex time-series data. The results demonstrate the strengths of our proposed model in the context of HT detection.

4.2 Evaluating the robustness of the proposed HT detection method against noise interference

Testing the robustness of the proposed HT detection method under noisy conditions is crucial, as hardware systems in real-world environments are exposed to various perturbations. Factors such as process variation, aging, and voltage level fluctuations can affect power consumption patterns, potentially masking the presence of HTs or generating signals that resemble HT-induced anomalies. To simulate these real-world conditions, white noise, denoted as $\mathcal{N}(0, \sigma^2)$, is introduced into the test signals, allowing us to assess the resilience of different HT detection methods under varying noise levels [46].

The robustness comparison across different methods is illustrated in Figs. 10, 11, 12, 13, and 14, which depict the variation in validation accuracy across different AES benchmark datasets (AES-T500, AES-T600, AES-T700, AES-T800, and AES-T1600) as noise standard deviation increases. In this section, in addition to the single-channel methods, we evaluate MTF Dual Channel for comparison, combining the MTF transformation with a raw signal reshaping branch to create a two-channel input for the CNN. The results show that the proposed dual-channel GAF-based method demonstrates superior robustness compared to the single-channel MTF and dual-channel MTF variants. This dual-channel MTF setup enhances feature extraction by using both temporal transition patterns (from MTF) and spatial information (from raw signal reshaping). While this approach improves robustness

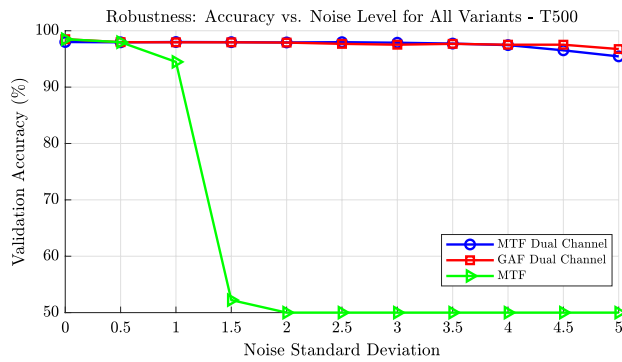


Fig. 10 Robustness analysis for the AES-T500 dataset: Validation accuracy (%) as a function of noise standard deviation for various models

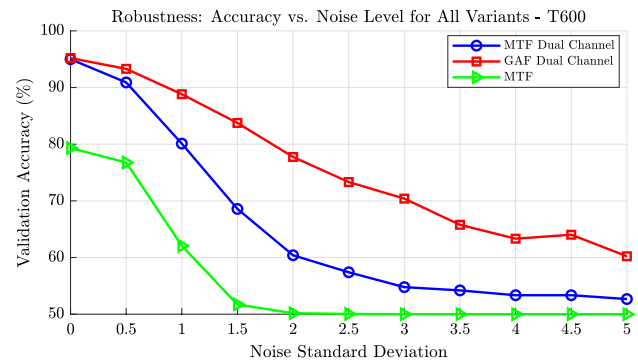


Fig. 11 Robustness analysis for the AES-T600 dataset: Validation accuracy (%) as a function of noise standard deviation for various models

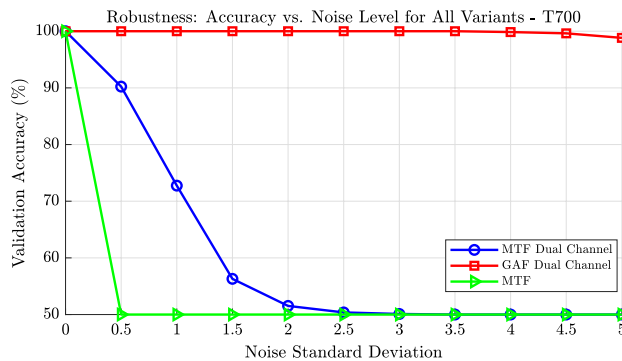


Fig. 12 Robustness analysis for the AES-T700 dataset: Validation accuracy (%) as a function of noise standard deviation for various models

against noise distortions compared to MTF alone, its performance still falls short of the dual-channel GAF-based method, indicating that GAF provides more resilient feature representations under noisy conditions.

For datasets such as AES-T500, AES-T700, and AES-T800, the proposed dual-channel GAF-based method sustains high accuracy, maintaining strong performance even under increased noise levels. This highlights the model's ability to extract and preserve discriminative features from both the GAF transformation and raw signal reshaping, making it more resilient to noise-induced distortions. In contrast, the single-channel MTF approach collapses early, showing little robustness beyond very low noise levels. The dual-channel MTF-based approach performs better than the single-channel methods but degrades more quickly than the dual-channel GAF variant, reinforcing that GAF-based transformations offer stronger feature representations for HT detection under noisy conditions.

For AES-T600 and AES-T1600, a more gradual decline in accuracy is observed across all methods as noise increases. These datasets pose additional challenges due to their complex and highly variable power consumption signals patterns, making it harder for models to maintain robust performance under perturbations. Despite this, dual-channel GAF outperforms all other methods, demonstrating its ability to handle the complexities of these datasets better than both the single-channel MTF and dual-channel MTF approaches. While the proposed method remains the most robust overall, its performance in extremely noisy conditions is influenced by the dataset's characteristics.

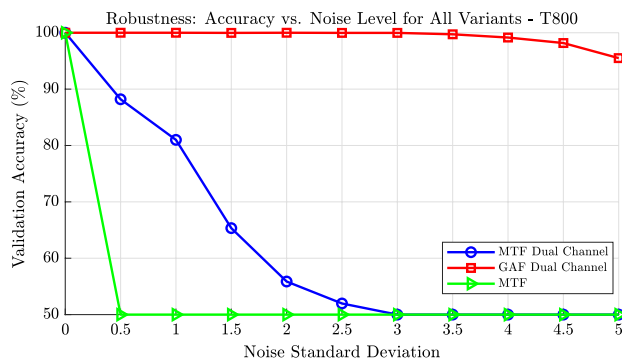


Fig. 13 Robustness analysis for the AES-T800 dataset: Validation accuracy (%) as a function of noise standard deviation for various models

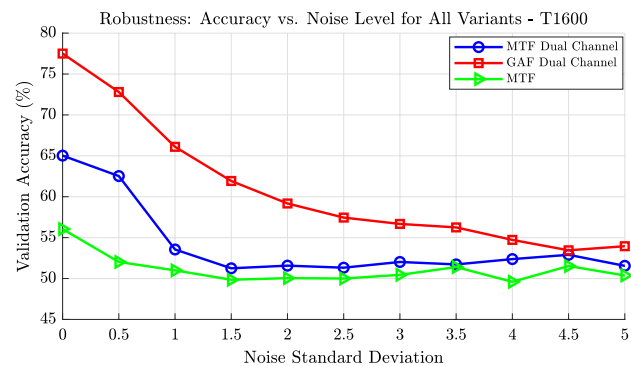


Fig. 14 Robustness analysis for the AES-T1600 dataset: Validation accuracy (%) as a function of noise standard deviation for various models

These findings emphasize that the dual-channel GAF-based method is the most effective in maintaining high detection accuracy across different noise levels and dataset complexities. The results validate the superior feature extraction and robustness capabilities of the proposed dual-channel GAF method, making it a practical and reliable choice for HT detection in real-world hardware environments where noise and perturbations are inevitable.

While Figs. 10–14 visualize robustness trends for the proposed dual-channel GAF and other CNN-based GAF/MTF variants, we further broaden the evaluation to include widely used baselines: an MLP on flattened raw input, a RawTrace-CNN on the raw waveform, an LSTM on the raw sequence, and an OC-SVM trained on raw features. All methods are tested under the same protocol, where Gaussian noise $\mathcal{N}(0, \sigma^2)$ is injected into the validation signals. To provide a concise summary across datasets and noise intensities, Table 3 reports the validation accuracy at representative noise levels $\sigma \in \{0.0, 2.5, 5.0\}$. These tabular results complement the figures by enabling direct, side-by-side comparison among a broader set of modeling approaches under identical noise conditions.

From Table 3, the proposed dual-channel GAF method consistently achieves the highest or near-highest accuracy across most datasets and noise levels, demonstrating its superior robustness to severe perturbations. In low-noise conditions ($\sigma = 0.0$), all deep learning-based methods achieve high accuracy, but as the noise increases to $\sigma = 2.5$ and $\sigma = 5.0$, the gap between the proposed approach and the baselines becomes more pronounced, particularly for the challenging AES-T600 and AES-T1600 datasets. The MLP baseline maintains moderate robustness but degrades more rapidly under heavy noise. The RawTrace-CNN shows competitive performance at low noise for some datasets but collapses toward chance-level accuracy for others when noise is high, indicating sensitivity to signal distortion. The LSTM maintains strong performance for some datasets (e.g., AES-T500) but exhibits significant drops for others, suggesting that temporal modeling alone does not guarantee robustness under severe perturbations. The OC-SVM, being a shallow unsupervised method, suffers the most from noise contamination and often converges to chance-level performance at higher noise intensities, since it lacks hierarchical feature extraction and relies directly on raw input distributions that are easily distorted by noise. Overall, these results reinforce that the dual-channel GAF design, which integrates complementary temporal and spatial feature representations, offers a more resilient foundation for hardware Trojan detection in noisy, real-world scenarios.

It is important to note that no explicit denoising or filtering techniques were applied during preprocessing. Instead, we intentionally retain the raw characteristics of the power traces, including intrinsic measurement noise, to promote end-to-end learning of noise-resilient features by the model itself. This design choice is based on the practical reality that real-world power side-channel data often contains non-trivial noise, and overly aggressive filtering may remove subtle Trojan-related patterns. As such, the proposed model is trained and validated on naturally noisy inputs, and its robustness is further assessed by injecting synthetic Gaussian noise into validation

Table 3 Validation accuracy of different methods across AES datasets at selected noise levels (σ)

Method	AES-T500			AES-T600			AES-T700			AES-T800			AES-T1600		
	$\sigma = 0.0$	$\sigma = 2.5$	$\sigma = 5.0$	$\sigma = 0.0$	$\sigma = 2.5$	$\sigma = 5.0$	$\sigma = 0.0$	$\sigma = 2.5$	$\sigma = 5.0$	$\sigma = 0.0$	$\sigma = 2.5$	$\sigma = 5.0$	$\sigma = 0.0$	$\sigma = 2.5$	$\sigma = 5.0$
Dual-channel GAF	0.985	0.985	0.967	0.974	0.733	0.602	1.000	1.000	0.988	1.000	1.000	0.997	0.775	0.609	0.535
MLP	0.850	0.850	0.830	0.655	0.600	0.510	1.000	0.910	0.851	1.000	0.934	0.844	0.705	0.566	0.527
ID CNN	0.980	0.980	0.500	0.932	0.640	0.512	1.000	0.500	0.500	1.000	0.500	0.500	0.687	0.520	0.500
LSTM	0.983	0.983	0.982	0.618	0.565	0.490	1.000	1.000	0.932	0.999	0.997	0.990	0.543	0.500	0.460
OC-SVM	0.897	0.500	0.500	0.497	0.482	0.487	0.902	0.500	0.500	0.927	0.500	0.500	0.502	0.501	0.490

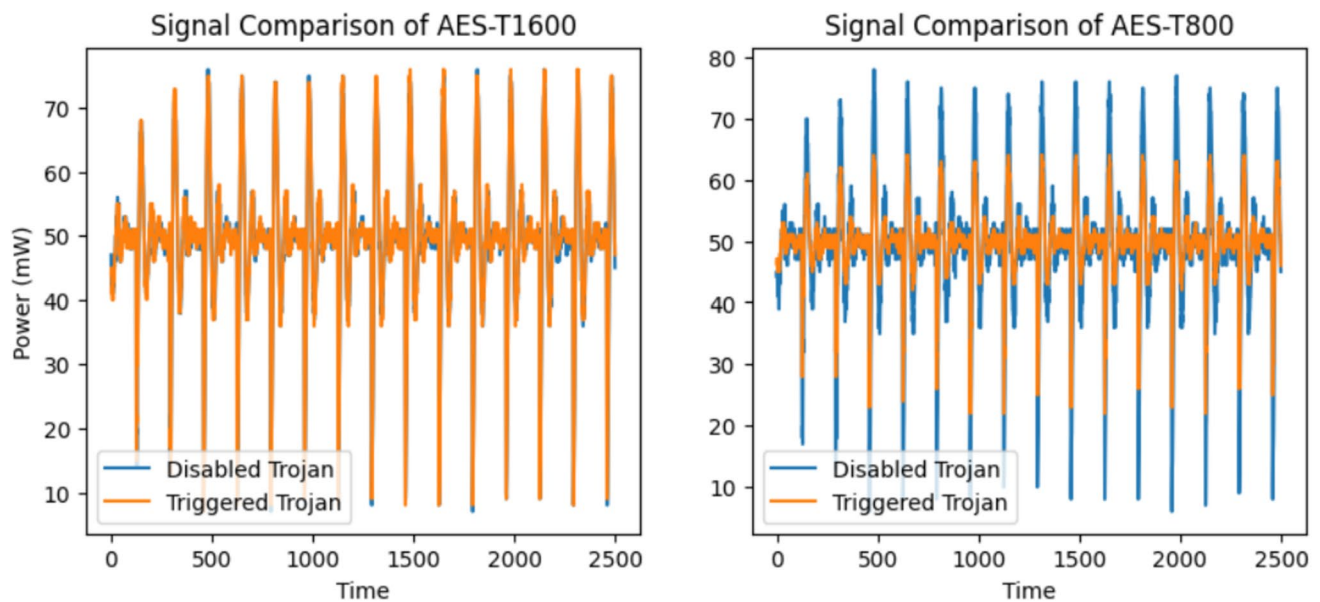


Fig. 15 Signal comparison of AES-T1600 (left) and AES-T800 (right)

signals. This evaluation pipeline reflects a realistic deployment scenario where external noise suppression is not always feasible.

4.3 Analyzing the effects of noise on signal discriminability for Trojan Detection

In addition to the effects of noise on the validation accuracy and loss of HT detection methods, it is crucial to understand how noise impacts the discriminability of signals in different benchmarks and why adding noise to test sets in some benchmark datasets affects HT detection performance. This section explores the influence of noise on the ability to distinguish between normal and HT-affected signals in selected AES benchmark datasets.

As explained earlier in equations (2) and (3), the presence of noise in the collected signals can significantly affect HT detection methods' performance. In particular, the discriminability of signals is a critical factor in determining the effectiveness of HT detection methods. When the difference between normal and HT-affected signals is similar to random noise, such as Gaussian noise, the signals become less distinguishable from each other [62]. As shown in the previous section, the impact of adding noise to a test set on signal discriminability can vary across different benchmarks. As shown in the previous section, the proposed time series imaging-based method for benchmarks like AES-T500, AES-T700, and AES-T800 exhibits remarkable stability in accuracy despite increasing noise levels, indicating that the detection method is robust at distinguishing between normal and HT-affected signals even in noisy environments. However, the proposed method for AES-T600 and AES-T1600 datasets is more sensitive to noise, leading to a decrease in discriminability and, consequently, a reduction in detection accuracy. Here, the AES-T1600 and AES-T800 datasets are selected to investigate noise impact on discriminability.

Figure 15 illustrates the comparison of power signals between disabled and triggered Trojan signals for the AES-T1600 and AES-T800 datasets. As depicted in the figure, the signals for the Trojan-triggered and Trojan-disabled states in the AES-T1600 dataset are very similar, while the difference is more apparent in the AES-T800 dataset. The difference signal can be obtained for two sample signals from AES-T1600 and AES-T800 as follows:

$$P_D^{T1600} = \mathcal{P}_T^{T1600} - \tilde{P}_T^{T1600} \quad (13)$$

$$P_D^{T800} = \mathcal{P}_T^{T800} - \tilde{P}_T^{T800} \quad (14)$$

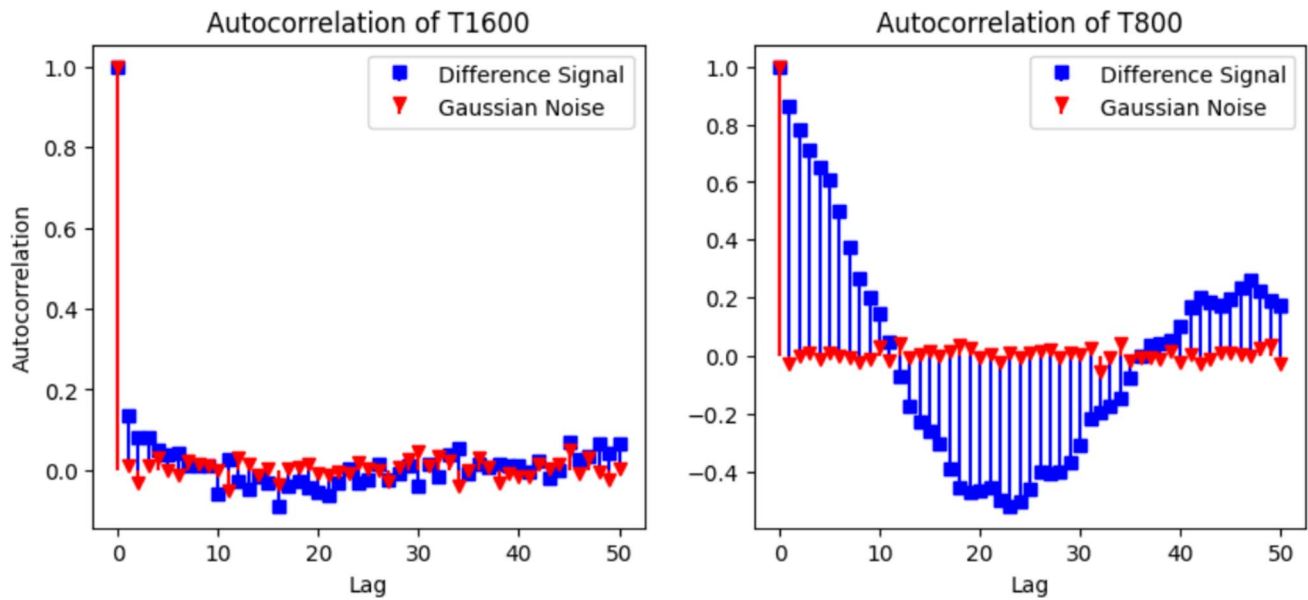


Fig. 16 Autocorrelation comparison of the difference signal and Gaussian noise for the AES-T1600 (left) and AES-T800 (right) benchmarks

where P_D^T represents the difference signal for a specific HT benchmark dataset, denoted by $T \in \{T1600, T800\}$, and P_T^T and $\tilde{P}_T^T$ are the power signals defined in equation (3). To further analyze these signals, the autocorrelation of the difference between triggered Trojan and disabled Trojan signals has been analyzed. Autocorrelation of a signal refers to the correlation of the signal with a delayed version of itself as a function of delay. It provides insight into the signal's repeating patterns or periodicity over time. High autocorrelation at a particular lag indicates a strong repetitive pattern at that time interval [63].

In our analysis, we hypothesize that if signal P_D^T exhibits behavior similar to Gaussian noise, then the perturbation introduced by the Trojan is effectively masked, making it harder for the detection model to distinguish it from random noise. To this aim, the autocorrelation of P_D^T is compared with the autocorrelation of Gaussian noise to demonstrate this similarity. To generate Gaussian noise, the mean μ_D^T and standard deviation σ_D^T of the difference signal P_D^T are obtained. Then, Gaussian noise is generated which can be modeled by the normal distribution $\mathcal{N}(\mu_D^T, (\sigma_D^T)^2)$. Following this, the autocorrelation of the difference signal P_D^T is calculated to determine the correlation of the signal with itself at different time lags. This step involves computing the correlation coefficient for each lag k , which can be formally expressed as:

$$R^T(k) = \frac{\sum_{i=1}^{N-k} (P_D^T(i) - \mu_D^T)(P_D^T(i+k) - \mu_D^T)}{\sum_{i=1}^N (P_D^T(i) - \mu_D^T)^2} \quad (15)$$

where N is the length of the time series signal.

The autocorrelation $R^T(k)$ of a difference signal P_D^T is compared with Gaussian noise in Fig. 16 for two sample data points from datasets of AES-T1600 and AES-T800. As shown in this figure, the autocorrelation P_D^{T1600} in AES-T1600 is similar to Gaussian noise, while for AES-T800, it is different. This is because the autocorrelation of P_D^{T1600} is high at lag 0, which indicates that the signal is perfectly correlated with itself but drops quickly to near zero for non-zero lags. However, the autocorrelation of P_D^{T800} in AES-T800 shows more fluctuation as the lag increases, indicating more variability compared to the AES-T1600 dataset. This pattern suggests that the differences between the triggered and disabled Trojan signals in AES-T1600 are random and unstructured, making it challenging for classifiers to identify features that reliably separate the Trojan-triggered and Trojan-disabled classes. Furthermore, this comparison evaluates the impact of noise on signal discriminability. As depicted in

Fig. 14, adding noise to the validation set deteriorates the accuracy of HT detection method for the AES-T1600 dataset. Therefore, a gradual decline in accuracy is observed for this dataset as the noise standard deviation increases. Specifically, when the impact of a Trojan attack resembles noise, as illustrated above, the introduction of additional noise to the test set likely compromises the ability to distinguish between the Trojan-triggered and Trojan-disabled classes. This makes it increasingly challenging to differentiate between these two classes and may lead to a reduction in detection accuracy by the HT detection method.

The following section summarizes the main contributions, key findings, and practical implications of this work, and suggests directions for future research.

4.4 Practical considerations on generality, complexity, and scalability

In addition to evaluating classification accuracy and robustness, it is important to discuss the practical aspects of the proposed framework. These include its ability to generalize beyond the AES benchmarks used in our experiments, as well as its computational complexity and runtime efficiency when deployed in realistic manufacturing environments.

4.4.1 Generality and transferability

Although the present evaluation is conducted on AES-based benchmarks from TrustHub, which are widely adopted in the hardware Trojan detection literature, the proposed dual-channel framework itself is not inherently limited to AES circuits. The combination of reshaped raw signals and GAF transformations provides a generic imaging pipeline for side-channel traces, independent of the specific cryptographic core. In practice, the same methodology can be applied to other cryptographic primitives such as RSA, SHA, or ECC. For instance, RSA and ECC, which are widely used public-key algorithms, exhibit side-channel leakage patterns related to modular arithmetic operations, while SHA as a hash function involves repetitive round-based computations that also leave distinct power and electromagnetic signatures. These operations generate time-series traces similar in structure to AES, making them compatible with the proposed imaging and CNN-based detection pipeline.

Beyond cryptographic circuits, the framework can also be extended to non-cryptographic hardware such as CPUs, SoCs, or IoT modules, provided that power or electromagnetic side-channel measurements are available. In such cases, only the data collection stage changes, while the CNN architecture and dual-channel transformation remain unchanged. Moreover, transfer learning strategies can further improve portability by reusing models trained on AES datasets and fine-tuning them for new hardware platforms with fewer training samples. This adaptability is particularly relevant for large-scale industrial settings where collecting extensive labeled datasets for each circuit is impractical. Consequently, the proposed approach has the potential to generalize beyond AES, supporting broader applicability across diverse hardware Trojan scenarios and strengthening its role in manufacturing-level security validation.

4.4.2 Complexity, runtime, and scalability

The proposed dual-channel CNN has approximately 1.87M trainable parameters and requires about 56.1 MFLOPs per inference on a $64 \times 64 \times 2$ input. On a workstation equipped with an NVIDIA Quadro RTX 6000 GPU, the per-sample inference latency was measured as 1.92 ms, while on CPU (Intel Xeon), the latency was 1.21 ms. These results indicate that the model can process hundreds of traces per second, supporting large-scale manufacturing testing. Training the model for 20 epochs on the AES benchmarks required only 5.3 min on the RTX 6000 GPU (approximately 15.93 s/epoch). The compact architecture and low FLOP count make the framework suitable for deployment on both high-performance GPUs and CPU-based test environments, demonstrating scalability to industrial screening scenarios. Since the transformations (reshaping and GAF) and convolutional operations are fixed for given input sizes, the overall complexity scales linearly with dataset size, enabling straightforward

parallelization across test stations or GPUs in production. This efficiency ensures that the framework can be practically integrated into industrial-level hardware verification workflows.

5 Conclusion

This paper has presented a novel approach for detecting Hardware Trojans (HTs) in chip devices, utilizing a dual-channel method combining Gramian Angular Field (GAF) transformations and deep learning techniques. By transforming side-channel signals into dual-channel image-like formats through GAF, we utilize the power of convolutional neural networks (CNNs) to effectively address the intricate task of HT detection. Our evaluation, using the Advanced Encryption Standard (AES) dataset from TrustHub and IEEE Dataport, has underscored the superior accuracy and classification precision of our method over existing techniques. The proposed dual-channel approach, exemplified by its robust feature extraction and generalization capabilities, achieved an accuracy of approximately 95.2% for AES-600 and 77.5% for AES-T1600, surpassing previous benchmarks. Moreover, we have evaluated the resilience of our dual-channel time-series imaging-based detection model against various noise levels representing real-world operational perturbations like process variation, aging, and voltage level variations. The remarkable stability of our model's accuracy, particularly for AES-T500, AES-T700, and AES-T800 benchmarks, even under escalating noise conditions, demonstrates exceptional robustness. For comparison, we also evaluated MTF Dual Channel, which combines MTF transformations with raw signal reshaping. While this method showed improved robustness over single-channel approaches, its performance was consistently lower than GAF Dual Channel, highlighting the superior resilience of GAF-based representations under noise perturbations. However, a decline in accuracy for AES-T600 and AES-T1600 with increasing noise suggests sensitivity to complex and uncertain power consumption patterns. In fact, the addition of noise, especially when it exhibits characteristics similar to HT-affected signals, can make it challenging for detection methods to distinguish between normal and HT-affected states. This is particularly evident in AES-T1600, where increasing noise levels lead to a marked decline in detection accuracy across all methods.

This study demonstrates the effectiveness of GAF transformations combined with deep learning for HT identification. Future work will focus on designing algorithms that are more robust to noise and developing methods that can generalize across a wider range of datasets, enabling the identification of different types of HT attacks with increased accuracy.

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